

SERVERLESS COMPUTING

Reducing Infrastructure Management in the Cloud

Function-as-a-Service: architecture, business advantages, and practical use cases

Guided by:

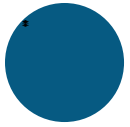
Dr. K. Sampath Kumar

Presented by:

Thamaraiselvi S
192511042
BE CSE (Dept)

What Is Serverless Computing?

A cloud execution model where the provider dynamically allocates and manages servers, letting developers deploy code without provisioning or maintaining infrastructure.



No server management

The cloud provider handles provisioning, patching, and capacity planning.



Automatic elastic scaling

Resources scale up and down instantly with real demand — no manual tuning.



Pay-per-execution pricing

Billing is based on actual compute time consumed, not idle reserved capacity.

Traditional vs Serverless

Application Code

Function Runtime

Provider-Managed Servers

Provider-Managed OS & Scaling

Dark block = your responsibility · Teal blocks = fully managed by the provider

Function-as-a-Service (FaaS)

FaaS is the execution model that powers serverless computing: applications are broken into small, single-purpose functions that run only in response to events.



Event-Triggered

Functions invoke only on demand
— an HTTP call, file upload,
queue message, or schedule.



Stateless Execution

Each invocation runs in an
isolated, ephemeral context with
no retained memory between
calls.



Granular Deployment

Code ships as small independent
functions instead of one large
monolithic application.

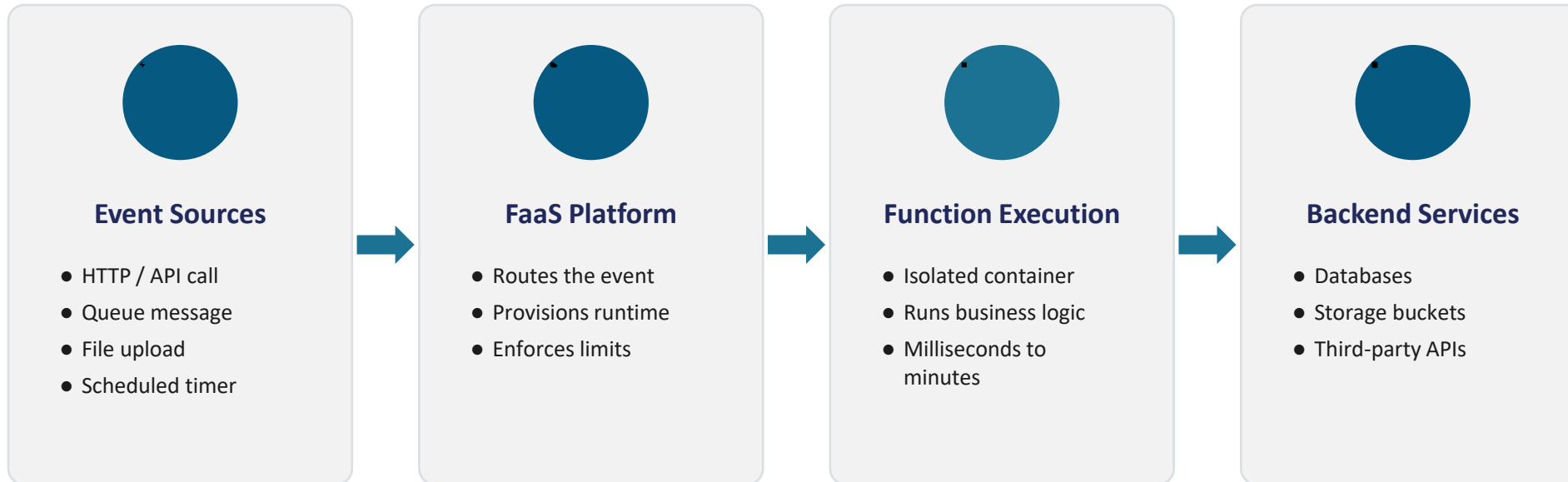


Instant Auto-Scaling

The platform spins up as many
parallel instances as incoming
events require.

FaaS Architecture

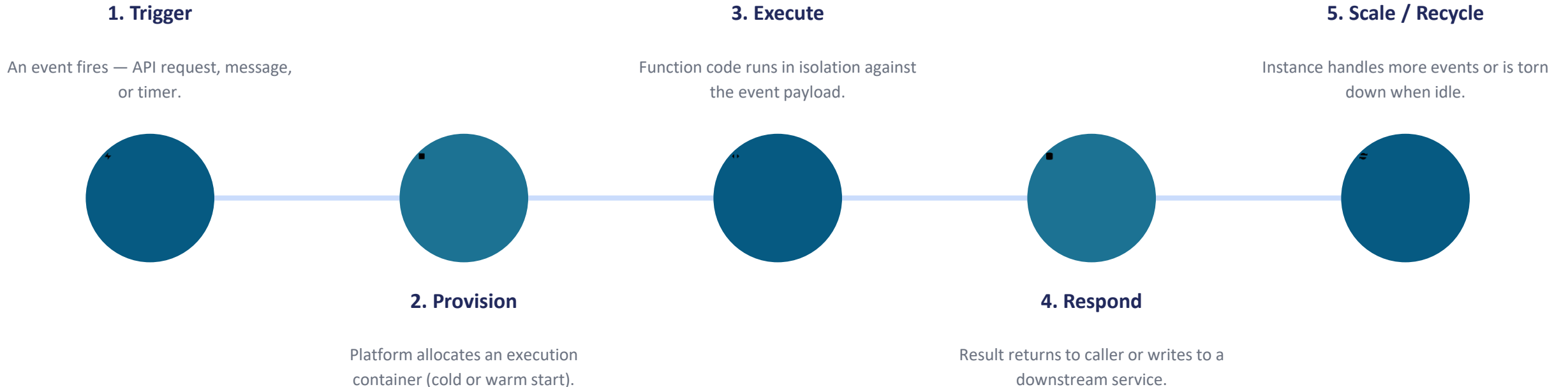
How an event travels from trigger to response inside a FaaS platform



Functions execute in parallel, on-demand instances — the platform manages capacity behind the scenes.

Function Execution Lifecycle

From triggering event to teardown — what happens on every invocation



Cold start: first request to a new instance takes longer to initialize than a **warm start** reusing an existing one.

Business Advantages

Why organizations adopt serverless FaaS beyond the technical elegance



Lower Total Cost

Pay only for execution time — no charge for idle servers.



Effortless Scaling

Handles traffic spikes automatically, without capacity planning.



Faster Time to Market

Teams ship features without managing servers or clusters.



Reduced Ops Burden

Patching, provisioning, and uptime shift to the provider.



Developer Focus

Engineers concentrate on business logic, not infrastructure.

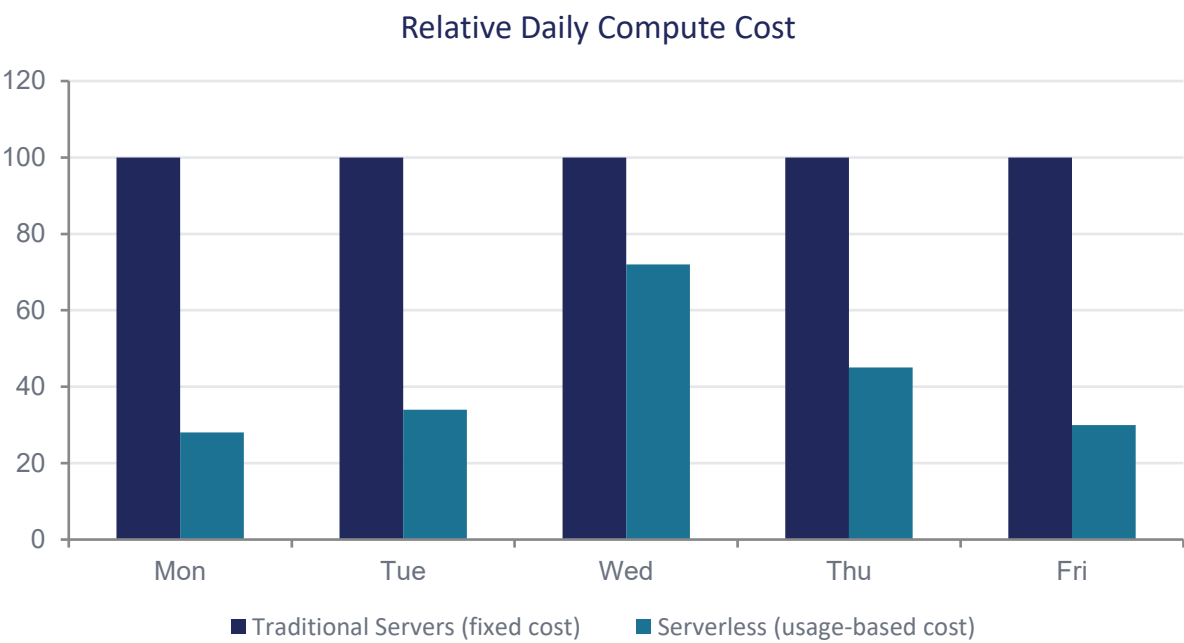


Built-In Resilience

Providers deliver multi-zone redundancy and failover by default.

The Serverless Cost Model

Provisioned servers bill for capacity; FaaS bills for consumption



Traditional capacity is provisioned for peak load and billed continuously; serverless cost tracks actual daily demand.

0

Servers to patch or manage

ms

Billing granularity for compute time

1000s

Concurrent function instances on demand

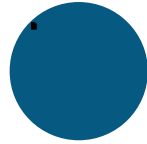
Practical Use Cases

Where FaaS delivers the clearest real-world impact



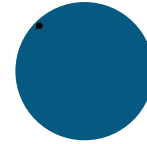
APIs & Microservices

Backend endpoints that scale per request without a standing server fleet.



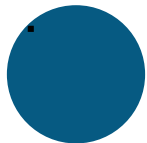
Data Processing & ETL

Transform files or streams the moment they land in storage.



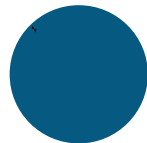
Chatbots & AI Backends

Lightweight inference and conversation logic invoked on demand.



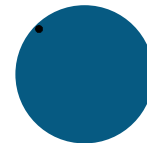
IoT Event Processing

Ingest and react to bursts of sensor data from connected devices.



Real-Time File Handling

Generate thumbnails, transcode media, or scan uploads instantly.



Scheduled & Cron Jobs

Recurring maintenance, reports, or cleanup with no dedicated host.

Leading FaaS Platforms

Major providers offering managed function execution today



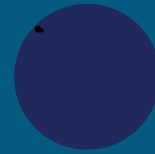
AWS Lambda

The original FaaS offering; deep integration across the AWS ecosystem.



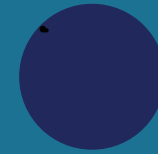
Azure Functions

Tight fit with Microsoft's cloud, DevOps, and enterprise identity tools.



Google Cloud Functions

Event-driven functions built around Google Cloud and Firebase.



Cloudflare Workers

Runs at the network edge for very low-latency global execution.

All four follow the same core pattern: upload code, define a trigger, let the platform run and scale it.

Challenges & Considerations

Serverless trades one set of problems for a different set



Cold Start Latency

New instances take extra time to initialize before serving the first request.



Vendor Lock-In

Deep use of provider-specific triggers and APIs can make migration costly.



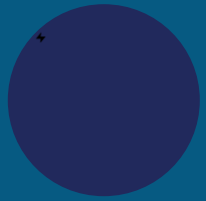
Execution Limits

Functions face strict timeouts, memory caps, and short-lived local storage.



Observability Complexity

Distributed, ephemeral functions make tracing and debugging harder.



Key Takeaways

- FaaS lets teams ship event-driven functions while the provider owns servers, scaling, and patching.
- The pay-per-execution model turns fixed infrastructure spend into variable, usage-aligned cost.
- It fits APIs, data pipelines, AI backends, IoT, and scheduled work particularly well.
- Cold starts, execution limits, and vendor lock-in remain real trade-offs to plan around.

Less infrastructure to manage. More time to build.

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Mevalurkuppam, Tamil Nadu, India 🇮🇳

Eee Department, Mevalurkuppam, Tamil Nadu 602105, India

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